Chapter ML:II (continued)

II. Machine Learning Basics

- Linear Regression
- Concept Learning: Search in Hypothesis Space
- Concept Learning: Version Space
- Evaluating Effectiveness

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Evaluating Effectiveness

Misclassification Rate

Definition 8 (True Misclassification Rate, True Error)

Let X be a set of feature vectors with a finite number of elements. Moreover, let C be a set of classes, let $y : X \to C$ be a classifier, and let $c(\mathbf{x})$ be the target concept to be learned. Then the true misclassification rate of $y(\mathbf{x})$, denoted by $Err^*(y)$, is defined as follows:

$$Err^*(y) = \frac{|\{\mathbf{x} \in X : y(\mathbf{x}) \neq c(\mathbf{x})\}|}{|X|}$$

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Evaluating Effectiveness

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Problem:

- Usually the *total function* $c(\mathbf{x})$ is unknown and hence *Err*$^*(y)$ is unknown.

Solution:

- Based on a set of examples D, estimation of upper and lower bounds for *Err*$^*(y)$ according to some sampling strategy.

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Remarks:

- Instead of the term “true misclassification rate” we will also use the term “true misclassification error” or simply “true error”.
- The unknown function $c(\mathbf{x})$ returns for a feature vector $\mathbf{x}$ its true class $c \in C$. I.e., “$c(\mathbf{x})$” and “c” designate the same, namely, the true class of $\mathbf{x}$ (if the $\mathbf{x}$ associated with c is clear).
- The English word “rate” can denote both the mathematical concept of a *flow quantity* (a change of a quantity per time unit) as well as the mathematical concept of a *portion*, a *percentage*, or a *ratio*, which has a stationary (= time-independent) semantics. Note that the latter semantics is meant here when talking about the misclassification rate.
- The German word „Rate“ is often (mis)used to denote the mathematical concept of a portion, a percentage, or a ratio. Taking a precise mathematical standpoint, the correct German words are „Anteil“ or „Quote“. I.e., a semantically correct translation of misclassification rate is „Missklassifikationsanteil“, and not „Missklassifikationsrate“.

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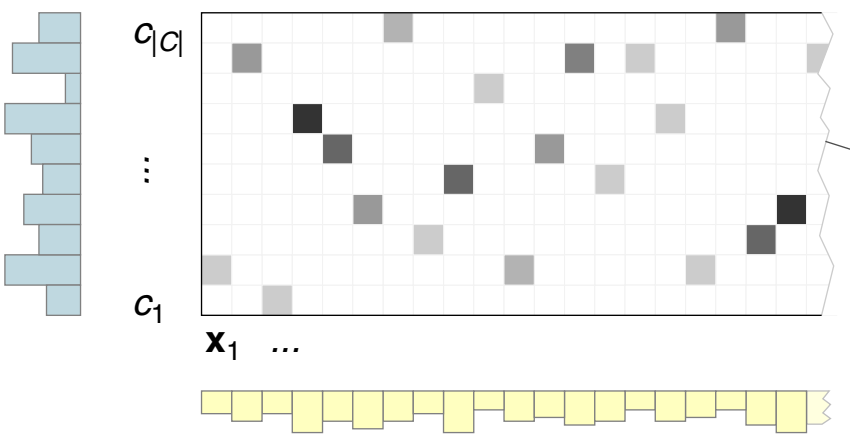
Evaluating Effectiveness

Misclassification Rate: Probabilistic Foundation

Let $\mathbf{X}$ be a feature space with a finite number of elements, C a set of classes, and P a probability measure on $\mathbf{X} \times C$. Then $p(\mathbf{x}, c)$, $p(\mathbf{x}, c) = P(\mathcal{X}{=}\mathbf{x}, \mathcal{C}{=}c)$, denotes the probability of the joint event (1) to observe the vector $\mathbf{x} \in \mathbf{X}$ and (2) that $\mathbf{x}$ belongs to class $c \in C$.

Illustration of P over $\mathbf{X} \times C$, $P \in [0; 1]$ (shading indicates magnitude) :



Joint probability $p(\mathbf{x}, c)$

(no label noise → classes unique)

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Evaluating Effectiveness

Misclassification Rate: Probabilistic Foundation (continued)

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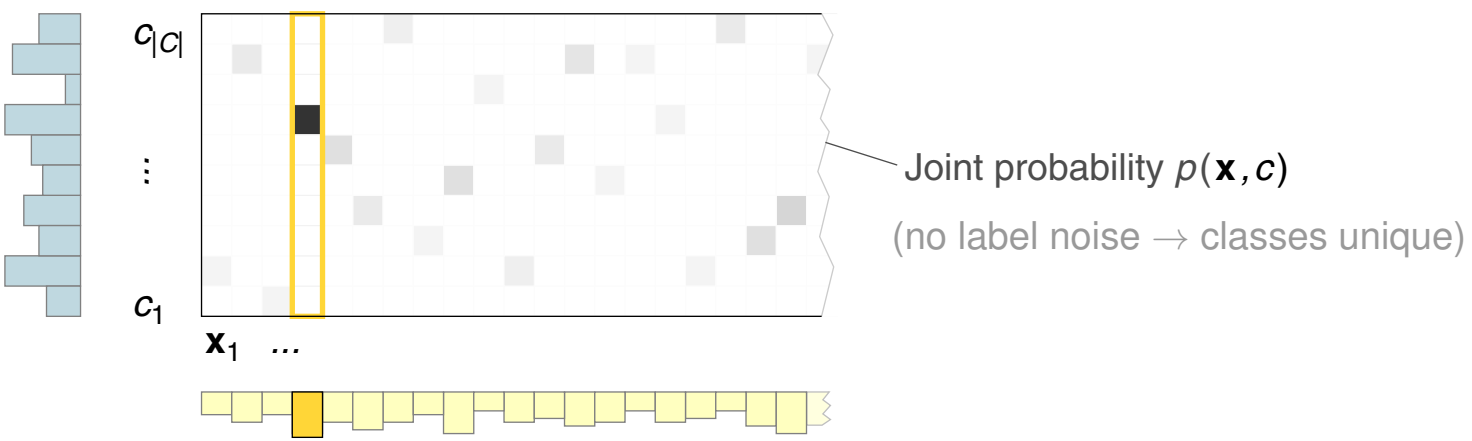

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Evaluating Effectiveness

Misclassification Rate: Probabilistic Foundation (continued)

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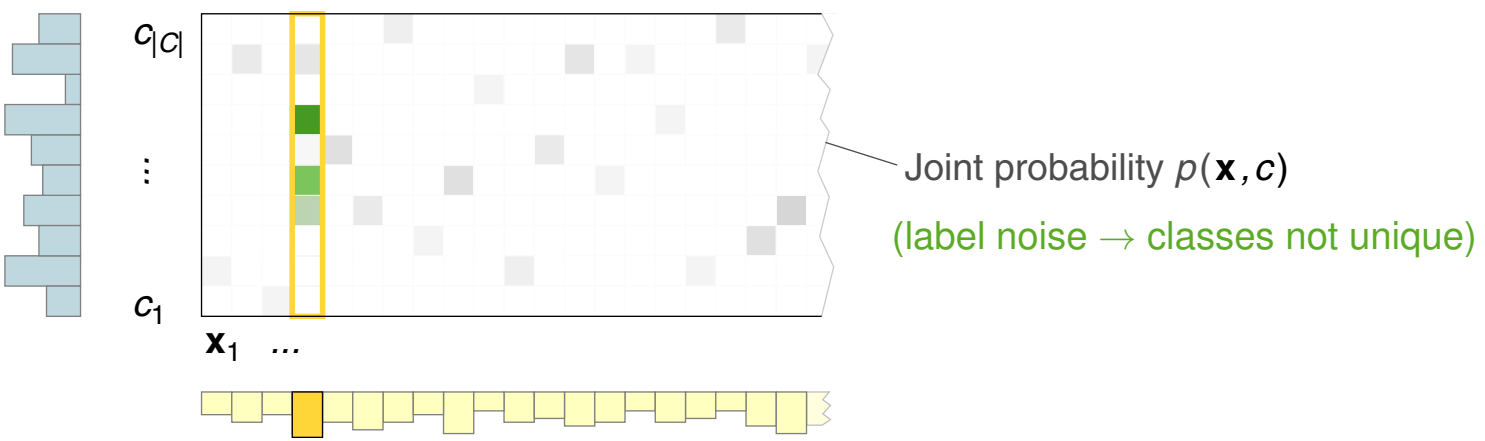

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Evaluating Effectiveness

Misclassification Rate: Probabilistic Foundation (continued)

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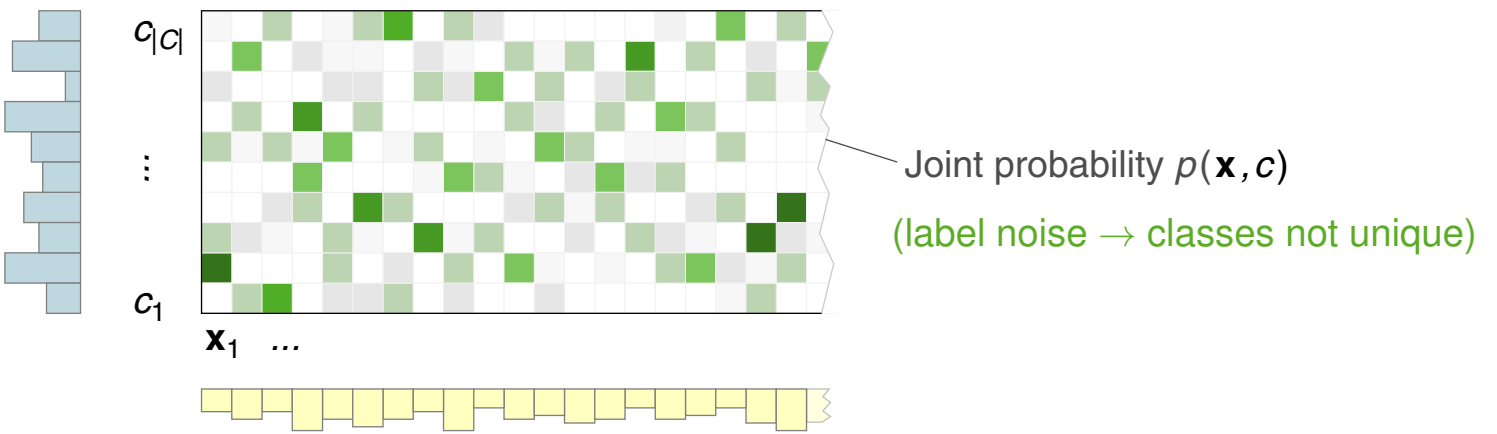

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Evaluating Effectiveness

Misclassification Rate: Probabilistic Foundation (continued)

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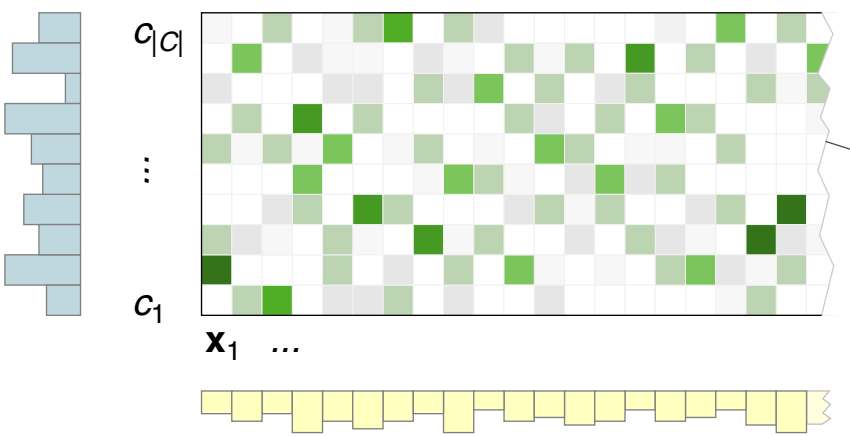

Joint probability $p(\mathbf{x}, c)$

(label noise → classes not unique)

$$\mathit{Err}^*(y) = \sum_{\mathbf{x} \in X} \sum_{c \in C} p(\mathbf{x}, c) \cdot I(y(\mathbf{x}), c), \quad \text{with } I(y(\mathbf{x}), c) = \begin{cases} 0 & \text{if } y(\mathbf{x}) = c \\ 1 & \text{otherwise} \end{cases}$$

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Remarks:

- Recall from Section Specification of Learning Tasks in part Introduction the difference between the following concepts, denoted by glyph variants of the same letter:
  x (single feature), $\mathbf{x}$ (feature vector), X (set of feature vectors), $\mathbf{X}$ (feature space), $\mathcal{X}$ (random variable whose realizations are feature vectors).

- $\mathcal{X}$ and $\mathcal{C}$ denote random variables with domains $\mathbf{X}$ and C respectively. $\mathbf{X}$ may not be restricted to a finite set, giving rise to probability density functions (with continuous random variables) in the place of the probability mass functions (with discrete random variables). The illustrations in a continuous setting remain basically unchanged, presupposing a sensible discretization of the feature space $\mathbf{X}$. [continuous setting]

- $\mathcal{X}$ models the fact that each observation procedure is governed by a probability distribution, rendering certain observations more likely than others. Keyword: prior probability of [observing some] $\mathbf{x}$.
  Note that the (multi)set X of feature vectors in the definition of the true misclassification rate is governed by the distribution of $\mathcal{X}$: Objects o that are more likely will induce the respective multiplicity of feature vectors $\mathbf{x}$ in X and hence are considered with the appropriate weight in $Err^*(y)$.

- $\mathcal{C}$ models the fact that in the real world the classification $c(\mathbf{x})$ of a feature vector $\mathbf{x}$ may not be deterministic. Reasons for a probabilistic class assignment include: an incomplete feature set, imprecision and random errors within the feature measuring process, lack of care during data acquisition. Keyword: label noise.

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Remarks: (continued)

- Since P is a probability measure (see section Probability Basics in part Statistical Learning), its argument is an event. Examples for events are "$\mathcal{X}{=}\mathbf{x}$", "$\mathcal{X}{=}\mathbf{x},\ \mathcal{C}{=}c$", or "$\mathcal{X}{=}\mathbf{x} \mid \mathcal{C}{=}c$".

- $p(\mathbf{x}, c)$, $p(\mathbf{x})$, $p(\mathbf{x} \mid c)$, etc. denote probabilty mass functions. A probabilty mass function, pmf, is defined for discrete random variables only. Its argument(s) is a (are) realization(s) of the respective random variable(s). The counterpart of $p()$ for continuous random variables is called probability density function, pdf, and is typically denoted by $f()$.

- Since $p(\mathbf{x}, c)$ is defined as $P(\mathcal{X}{=}\mathbf{x}, \mathcal{C}{=}c)$ the two expressions can be used interchangeably in most places.

- Let A and B denote two events, e.g., $A =$ "$\mathcal{X}{=}\mathbf{x}_9$" and $B =$ "$\mathcal{C}{=}c_3$". Then the following expressions are equivalent notations for the probability of the joint event: $P(A, B)$, $P(A \text{ and } B)$, $P(A \wedge B)$.

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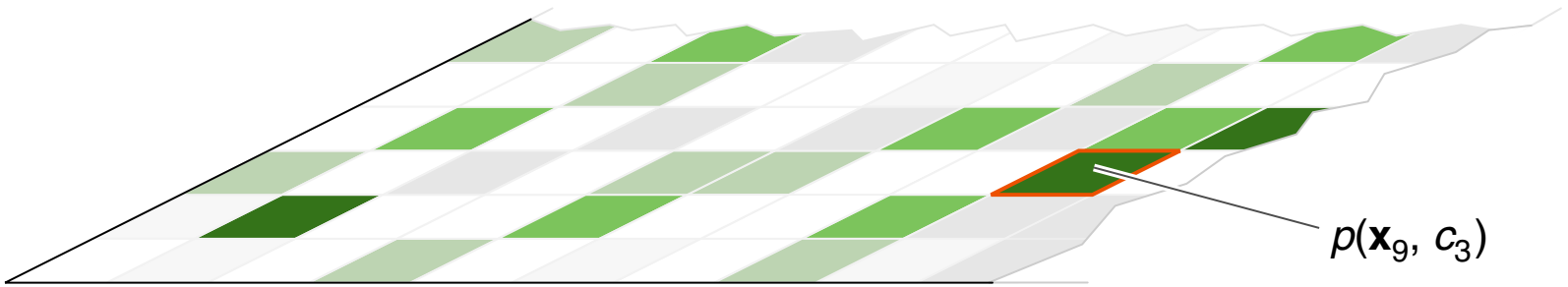
Evaluating Effectiveness

Illustration 1: Generic Feature and Class Distributions [Wikipedia]

Joint probability $p(\mathbf{x}, c) = P(\mathcal{X}{=}\mathbf{x}, \mathcal{C}{=}c)$:



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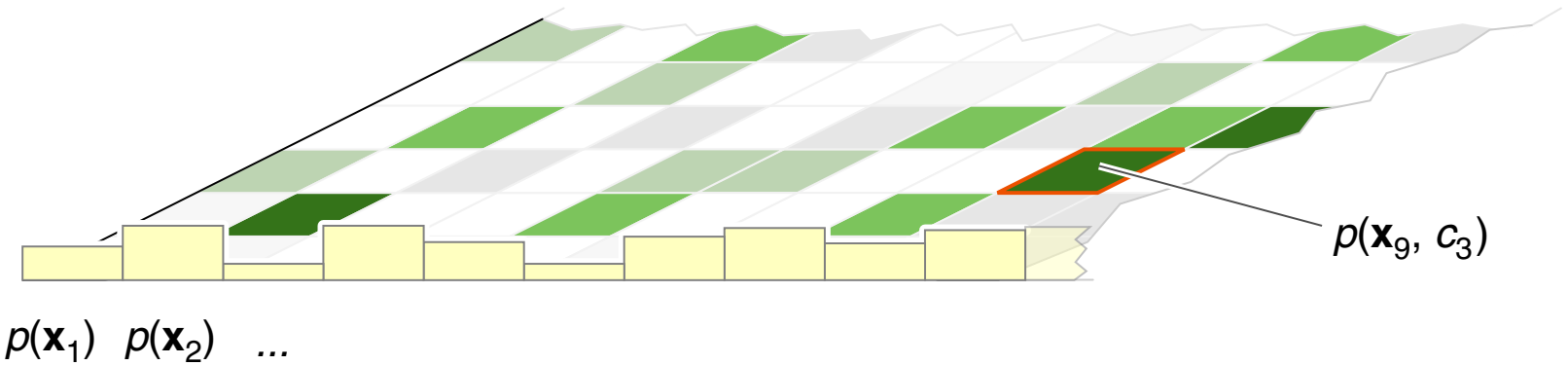
Evaluating Effectiveness

Illustration 1: Generic Feature and Class Distributions [Wikipedia] (continued)

Marginal probability $p(\mathbf{x}) = P(\mathcal{X}{=}\mathbf{x})$:



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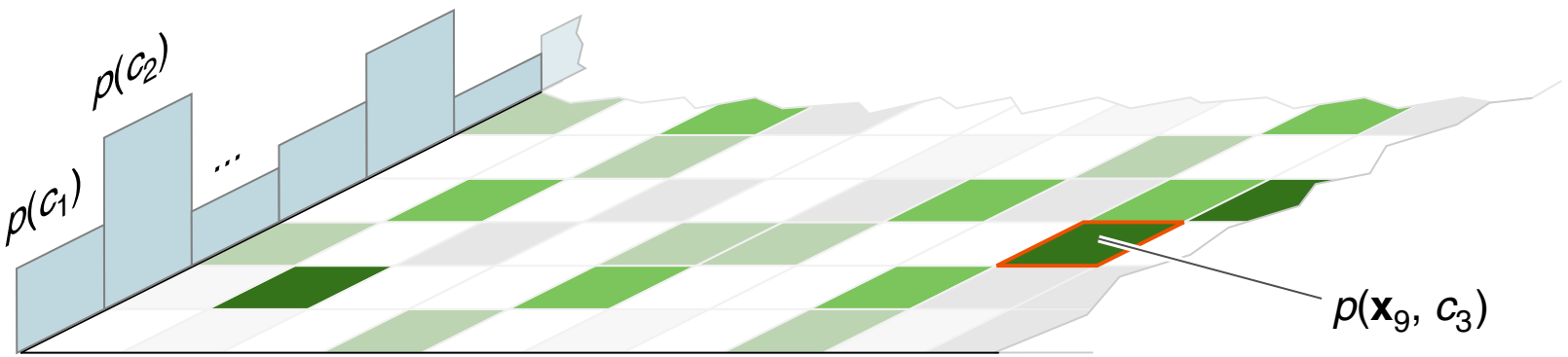
Evaluating Effectiveness

Illustration 1: Generic Feature and Class Distributions [Wikipedia] (continued)

Marginal probability $p(c) = P(\mathcal{C}{=}c)$:



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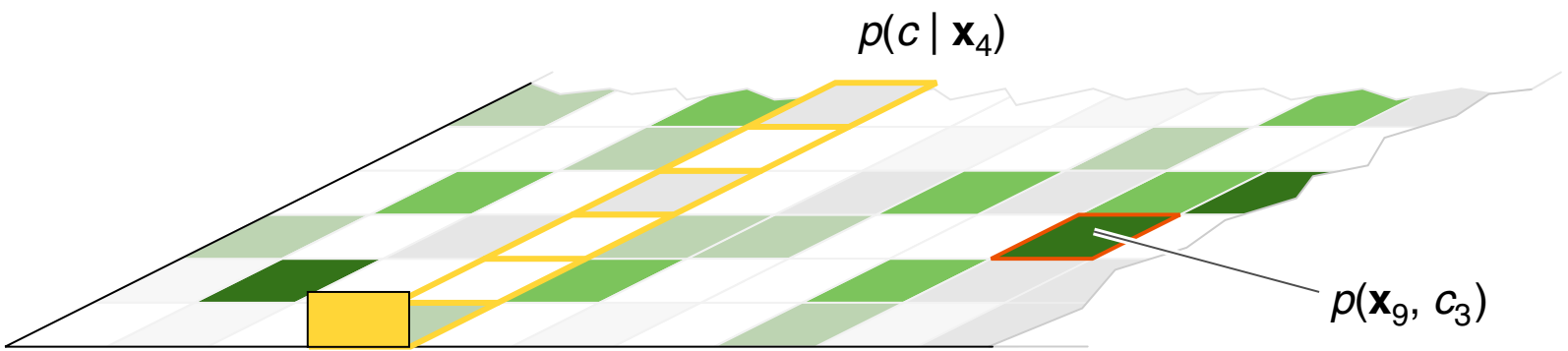
Evaluating Effectiveness

Illustration 1: Generic Feature and Class Distributions [Wikipedia] (continued)

Distribution of the classes c under feature vector $\mathbf{x}_4$,
denoted by $p(c \mid \mathbf{x}_4) = P(\mathcal{C}{=}c \mid \mathcal{X}{=}\mathbf{x}_4) \;\equiv\; P_{\mathcal{X}=\mathbf{x}_4}(\mathcal{C}{=}c)$:



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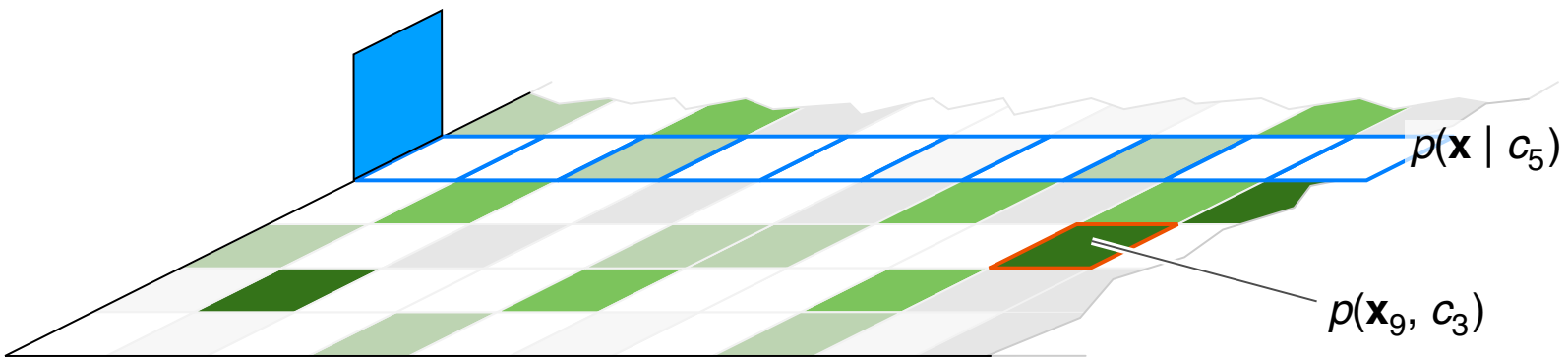
Evaluating Effectiveness

Illustration 1: Generic Feature and Class Distributions [Wikipedia] (continued)

Distribution of the feature vectors $\mathbf{x}$ und class c_5,
denoted by $p(\mathbf{x} \mid c_5) = P(\mathcal{X}{=}\mathbf{x} \mid \mathcal{C}{=}c_5) \;\equiv\; P_{\mathcal{C}=c_5}(\mathcal{X}{=}\mathbf{x})$:



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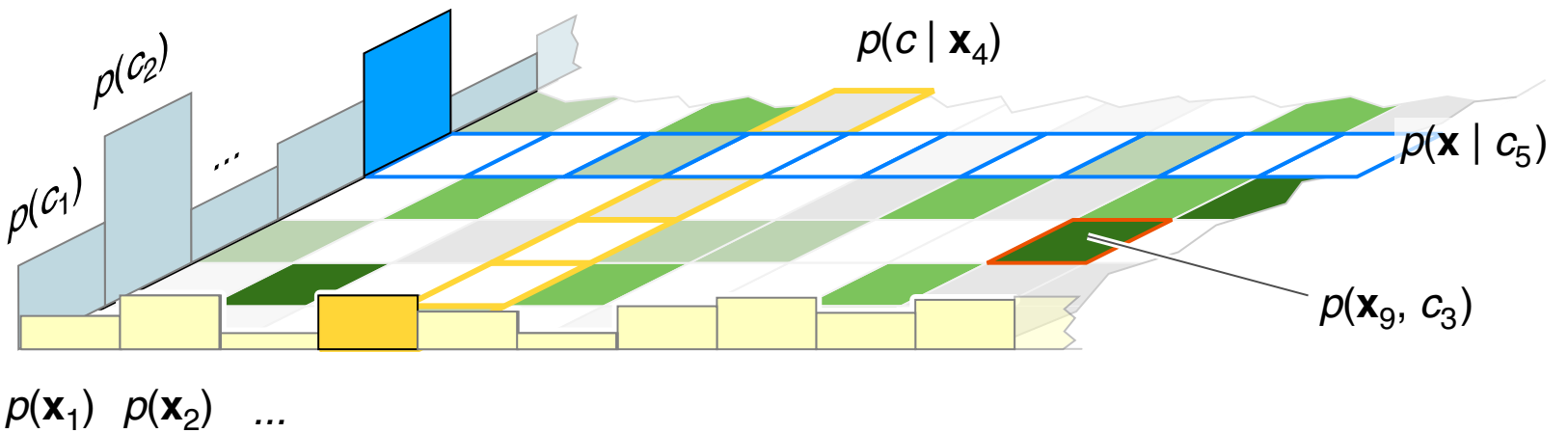
Evaluating Effectiveness

Illustration 1: Generic Feature and Class Distributions [Wikipedia] (continued)

Overview:



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Remarks:

- $p(c \mid \mathbf{x}) = P(\mathcal{X}{=}\mathbf{x}, \mathcal{C}{=}c)/P(\mathcal{X}{=}\mathbf{x}) = P(\mathcal{C}{=}c \mid \mathcal{X}{=}\mathbf{x}) \;\equiv\; P_{\mathcal{X}=\mathbf{x}}(\mathcal{C}{=}c)$

  $p(c \mid \mathbf{x})$ is called "(feature-)conditional class probability" function.

  In the illustration: Summation over the $c \in C$ of the fourth column yields the marginal probability $p(\mathbf{x}_4) = P(\mathcal{X}{=}\mathbf{x}_4)$. $p(c \mid \mathbf{x}_4)$ is the distribution of the c (consider the column) under feature vector $\mathbf{x}_4$ (= having normalized by $p(\mathbf{x}_4)$), i.e., $p(\mathbf{x}_4, c)/p(\mathbf{x}_4)$.

- $p(\mathbf{x} \mid c) = P(\mathcal{X}{=}\mathbf{x}, \mathcal{C}{=}c)/P(\mathcal{C}{=}c) = P(\mathcal{X}{=}\mathbf{x} \mid \mathcal{C}{=}c) \;\equiv\; P_{\mathcal{C}=c}(\mathcal{X}{=}\mathbf{x})$

  $p(\mathbf{x} \mid c)$ is called "class-conditional (feature) probability [density]" function.

  In the illustration: Summation / integration over the $\mathbf{x} \in X$ of the fifth row yields the marginal probability $p(c_5) = P(\mathcal{C}{=}c_5)$. $p(\mathbf{x} \mid c_5)$ is the distribution of the $\mathbf{x}$ (consider the row) under class c_5 (= having normalized by $p(c_5)$), i.e., $p(\mathbf{x}, c_5)/p(c_5)$.

- $P(\mathcal{X}{=}\mathbf{x}, \mathcal{C}{=}c) = P(\mathcal{C}{=}c, \mathcal{X}{=}\mathbf{x}) = P(\mathcal{C}{=}c \mid \mathcal{X}{=}\mathbf{x}) \cdot P(\mathcal{X}{=}\mathbf{x})$, where $P(\mathcal{X}{=}\mathbf{x})$ is the prior probability for event $\mathcal{X}{=}\mathbf{x}$, and $P(\mathcal{C}{=}c \mid \mathcal{X}{=}\mathbf{x})$ is the probability for event $\mathcal{C}{=}c$ given event $\mathcal{X}{=}\mathbf{x}$.

  Likewise, $P(\mathcal{X}{=}\mathbf{x}, \mathcal{C}{=}c) = P(\mathcal{X}{=}\mathbf{x} \mid \mathcal{C}{=}c) \cdot P(\mathcal{C}{=}c)$, where $P(\mathcal{C}{=}c)$ is the prior probability for event $\mathcal{C}{=}c$, and $P(\mathcal{X}{=}\mathbf{x} \mid \mathcal{C}{=}c)$ is the probability for event $\mathcal{X}{=}\mathbf{x}$ given event $\mathcal{C}{=}c$.

- Let the events $\mathcal{X}{=}\mathbf{x}$ and $\mathcal{C}{=}c$ have occurred, and, let $\mathbf{x}$ be known and c be unknown. Then, $P(\mathcal{X}{=}\mathbf{x} \mid \mathcal{C}{=}c)$ is called *likelihood* (for event $\mathcal{X}{=}\mathbf{x}$ given event $\mathcal{C}{=}c$). [Mathworld]

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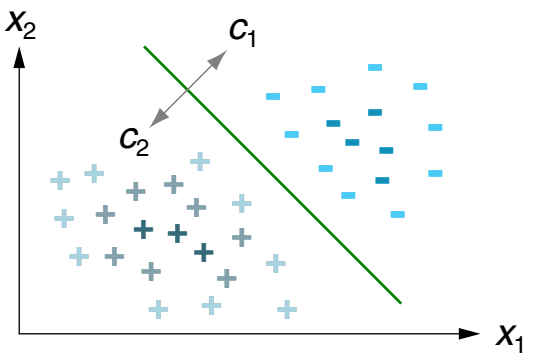
Evaluating Effectiveness

Illustration 2: Feature Distribution given Linearly Separable Classes



$$\mathbf{X} = \left\{ \underbrace{\begin{pmatrix} x_{1_1} \\ x_{1_2} \end{pmatrix}}_{\mathbf{x}_1}, \underbrace{\begin{pmatrix} x_{2_1} \\ x_{2_2} \end{pmatrix}}_{\mathbf{x}_2}, \underbrace{\ldots}_{\ldots} \right\} \subset \mathbf{R}^2$$

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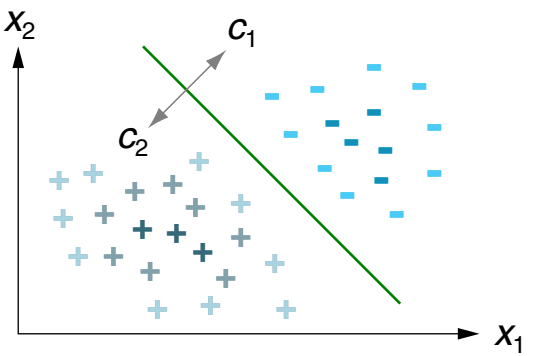
Evaluating Effectiveness

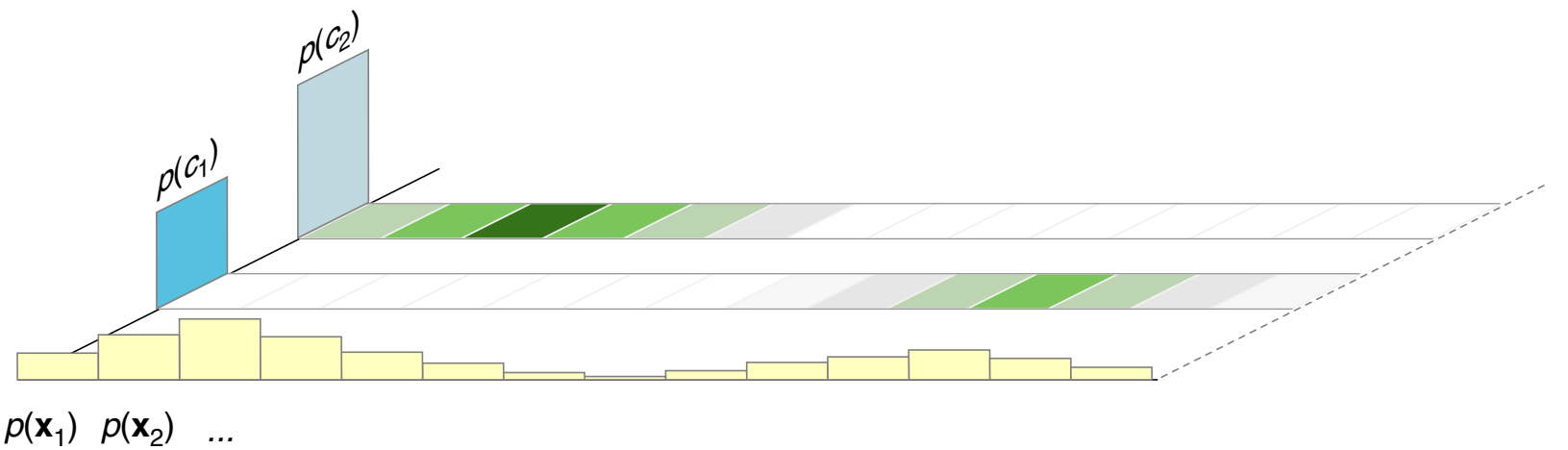
Illustration 2: Feature Distribution given Linearly Separable Classes (continued)



$$\mathbf{X} = \left\{ \begin{pmatrix} x_{1_1} \\ x_{1_2} \end{pmatrix}, \begin{pmatrix} x_{2_1} \\ x_{2_2} \end{pmatrix}, \ldots \right\} \subset \mathbf{R}^2$$

$$\mathbf{x}_1 \qquad \mathbf{x}_2 \qquad \ldots$$

Joint and marginal probability functions $p(\mathbf{x}, c)$, $p(\mathbf{x})$, and $p(c)$:



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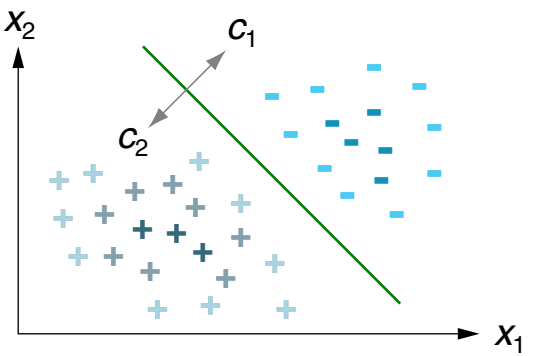
Evaluating Effectiveness

Illustration 2: Feature Distribution given Linearly Separable Classes (continued)



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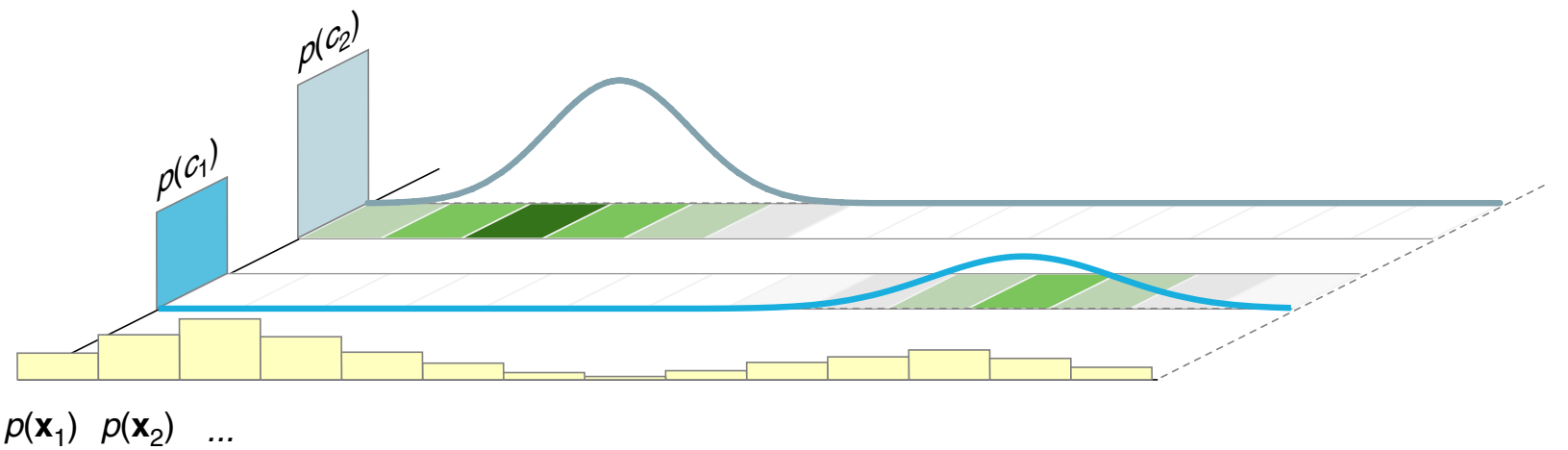
$$\mathbf{x}_1 \qquad \mathbf{x}_2 \qquad \ldots$$

Joint and marginal probability functions $p(\mathbf{x}, c)$, $p(\mathbf{x})$, and $p(c)$:



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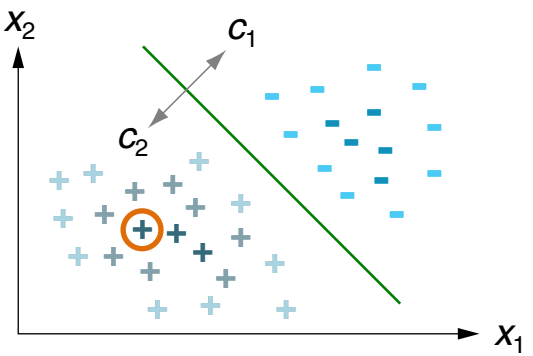
Evaluating Effectiveness

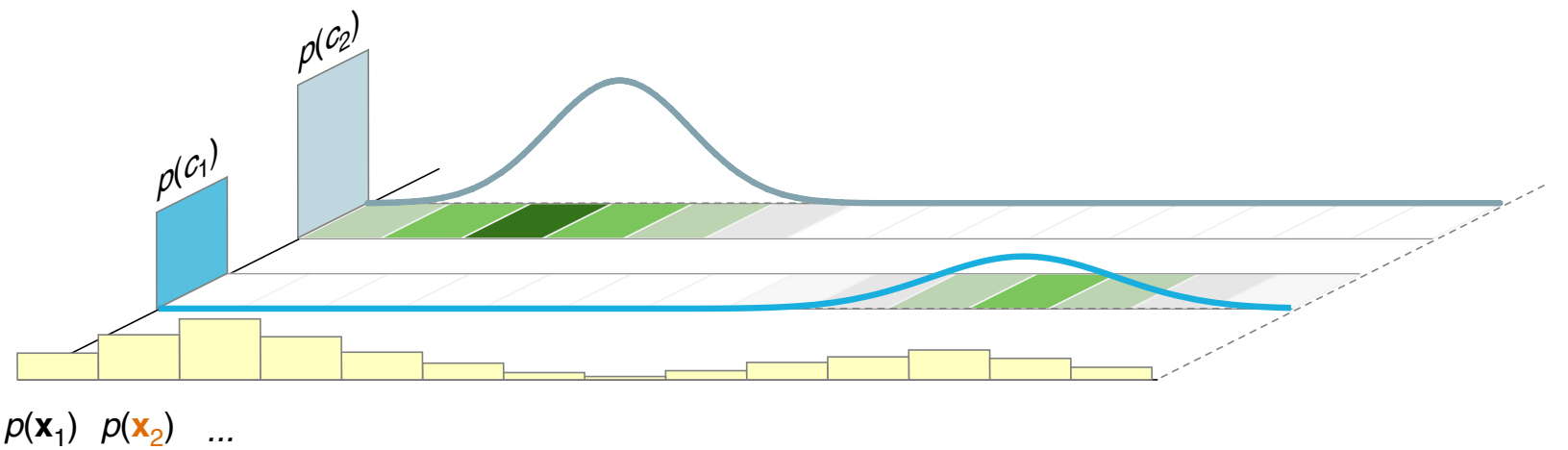
Illustration 2: Feature Distribution given Linearly Separable Classes (continued)



$$\mathbf{X} = \left\{ \begin{pmatrix} x_{1_1} \\ x_{1_2} \end{pmatrix}, \begin{pmatrix} x_{2_1} \\ x_{2_2} \end{pmatrix}, \ldots \right\} \subset \mathbf{R}^2$$

$$\mathbf{x}_1 \qquad \mathbf{x}_2 \qquad \ldots$$

Joint and marginal probability functions $p(\mathbf{x}, c)$, $p(\mathbf{x})$, and $p(c)$:



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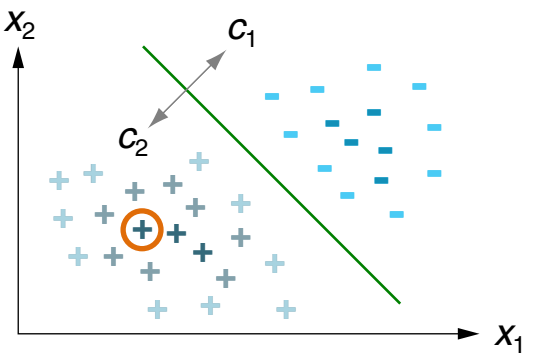
Evaluating Effectiveness

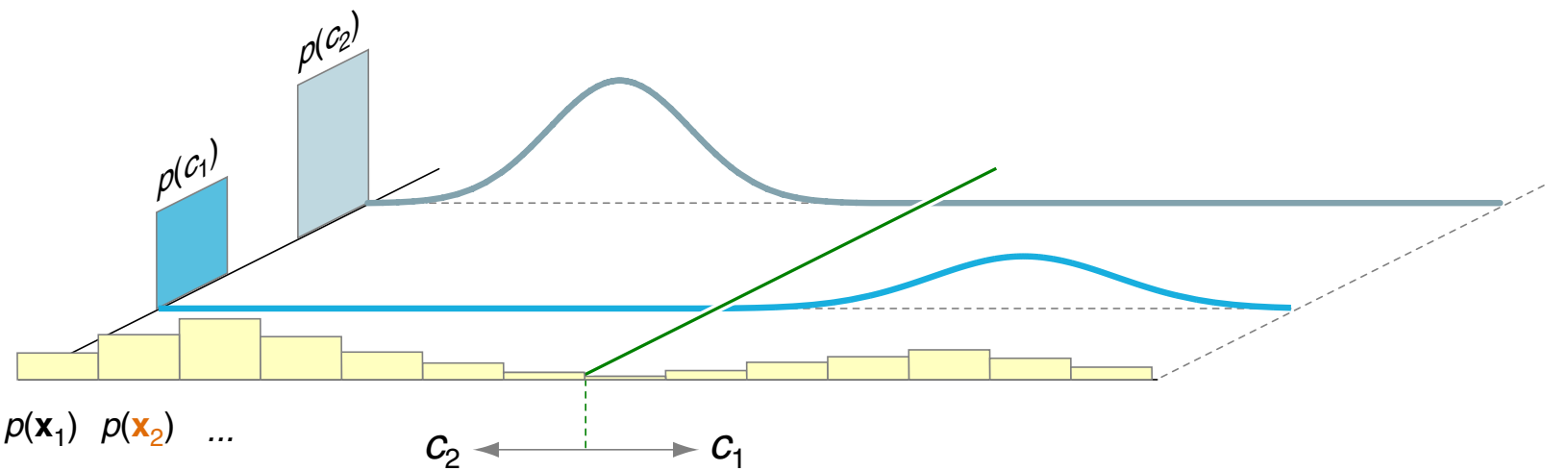
Illustration 2: Feature Distribution given Linearly Separable Classes (continued)



$$\mathbf{X} = \left\{ \begin{pmatrix} x_{1_1} \\ x_{1_2} \end{pmatrix}, \begin{pmatrix} x_{2_1} \\ x_{2_2} \end{pmatrix}, \ldots \right\} \subset \mathbf{R}^2$$

$$\mathbf{x}_1 \qquad \mathbf{x}_2 \qquad \ldots$$

Optimum hyperplane classifier:



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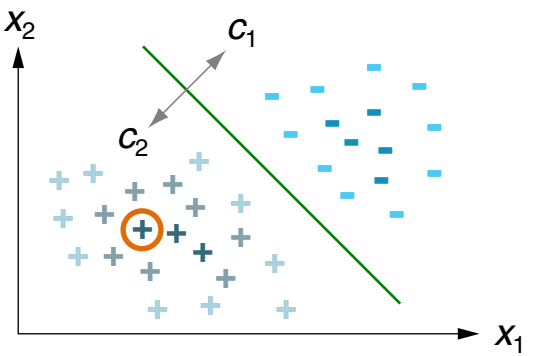
Evaluating Effectiveness

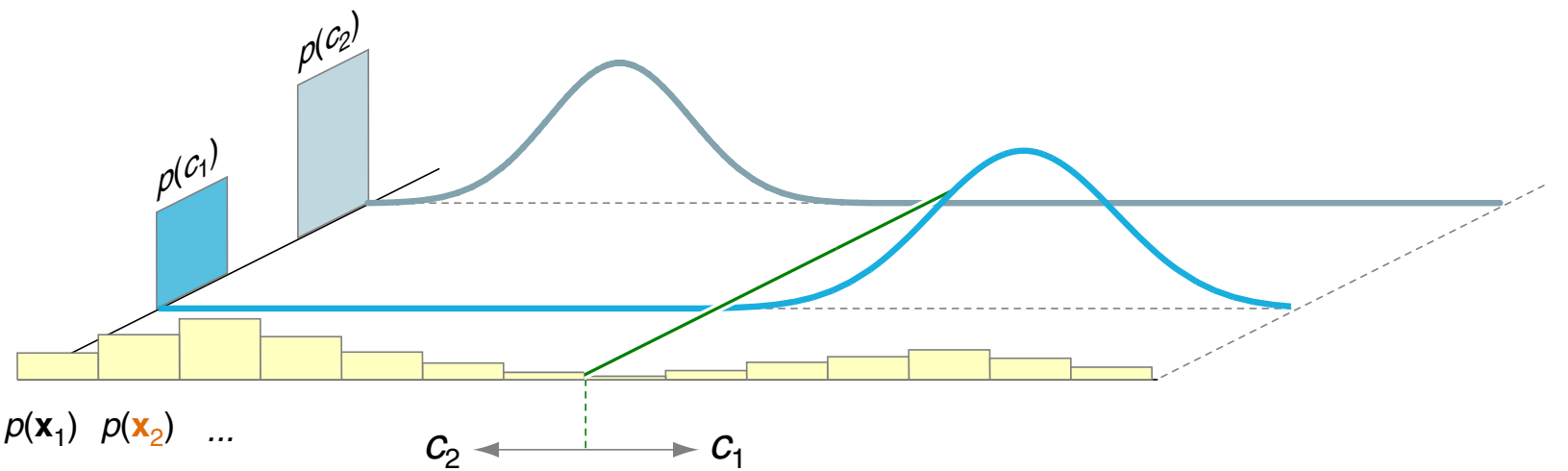
Illustration 2: Feature Distribution given Linearly Separable Classes (continued)



$$\mathbf{X} = \left\{ \begin{pmatrix} x_{1_1} \\ x_{1_2} \end{pmatrix}, \begin{pmatrix} x_{2_1} \\ x_{2_2} \end{pmatrix}, \ldots \right\} \subset \mathbf{R}^2$$

$$\mathbf{x}_1 \qquad \mathbf{x}_2 \qquad \ldots$$

Class-conditional probability functions $p(\mathbf{x} \mid c_1)$ and $p(\mathbf{x} \mid c_2)$:



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Remarks:

- Actually, the two figures illustrate the discriminative approach (top) and the generative approach (bottom) to classification. See section Elements of Machine Learning in part Introduction.
- Linear separability requires a feature space with a particular structure. Here we assume that the feature space is a vector space over the scalar field of real numbers $\mathbb{R}$, equipped with the dot product.

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Evaluating Effectiveness

Estimating Error Bounds

Classifier training:

- $D = \{(\mathbf{x}_1, c(\mathbf{x}_1)), \ldots, (\mathbf{x}_n, c(\mathbf{x}_n))\} \subseteq X \times C$ is a set of examples.
- $y(\mathbf{x})$ is the classifier trained on D.
- The true error $\mathit{Err}^*(y)$ measures the performance of $y(\mathbf{x})$ on X ("in the wild").
- What can be said about the true error $\mathit{Err}^*(y)$ of $y(\mathbf{x})$?

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Evaluating Effectiveness

Estimating Error Bounds

Classifier training:

- $D = \{(\mathbf{x}_1, c(\mathbf{x}_1)), \ldots, (\mathbf{x}_n, c(\mathbf{x}_n))\} \subseteq X \times C$ is a set of examples.
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The following relations typically hold:

Training error		Cross-validation error		Holdout error
$Err_{tr}(y)$	$<$	$Err_{cv}(y, k)$	$<$	$Err(y, D_{test})$

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Evaluating Effectiveness

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The following relations typically hold:

Training error		Cross-validation error		Holdout error		True error
$Err_{tr}(y)$	$<$	$Err_{cv}(y, k)$	$<$	$Err(y, D_{test})$	$<$	
						$Err^*(y)$

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Remarks:

- Relating the true error $Err^*(y)$ to the aforementioned error assessments $Err_{tr}(y)$, $Err_{cv}(y,k)$, and $Err(y, D_{test})$ is not straightforward but requires an in-depth analysis of the sampling strategy, the sample size D, and the set X of feature vectors, among others.
- With $D = X$ the true error $Err^*(y)$ becomes the training error $Err_{tr}(y)$.

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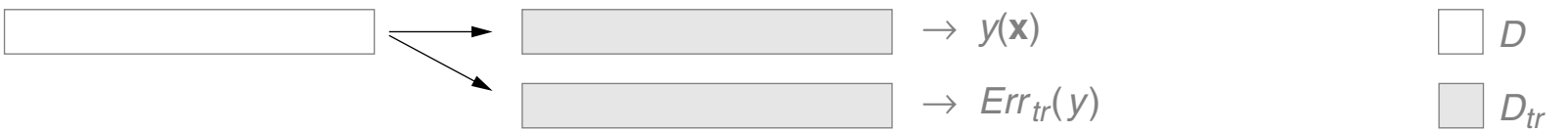
Evaluating Effectiveness

Training Error



Setting:

- No test set.
- $y(\mathbf{x})$ is the classifier trained on $D_{tr} = D$.

Training error of $y(\mathbf{x})$:

- $$Err_{tr}(y) = \frac{|\{(\mathbf{x}, c(\mathbf{x})) \in D_{tr} : y(\mathbf{x}) \neq c(\mathbf{x})\}|}{|D_{tr}|}$$

  $=$ misclassification rate of $y(\mathbf{x})$ on the training set.

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Remarks:

- $\mathit{Err}_{tr}(y)$ is based on examples that are also exploited to train $y(\mathbf{x})$. It hence quantifies the *memorization* capability but not the *generalization* capability of $y(\mathbf{x})$.

- $\mathit{Err}_{tr}(y)$ is an optimistic estimation, i.e., it is constantly lower compared to the (unknown) true error $\mathit{Err}^*(y)$.

- Note that the above issues relate to the meaningfulness of $\mathit{Err}_{tr}(y)$ as an error estimate—and not to the classifier $y(\mathbf{x})$.
  Obviously, to get the maximum out of the data when training $y(\mathbf{x})$, D must be exploited completely: A classifier $y(\mathbf{x})$ trained on D will on average outperform any classifier $y'(\mathbf{x})$ trained on a subset of D. Hence, the error of $y'(\mathbf{x})$ may be considered as a lower bound for the (unknown) error of $y(\mathbf{x})$.

- Regarding notation: We will us the prime symbol »'« to indicate whether a classifier is trained by withholding a test set. E.g., $y'(\mathbf{x})$ and $y'_i(\mathbf{x})$ denote classifiers trained by withholding the test sets D_{test} and $D_{test\,i}$ respectively.

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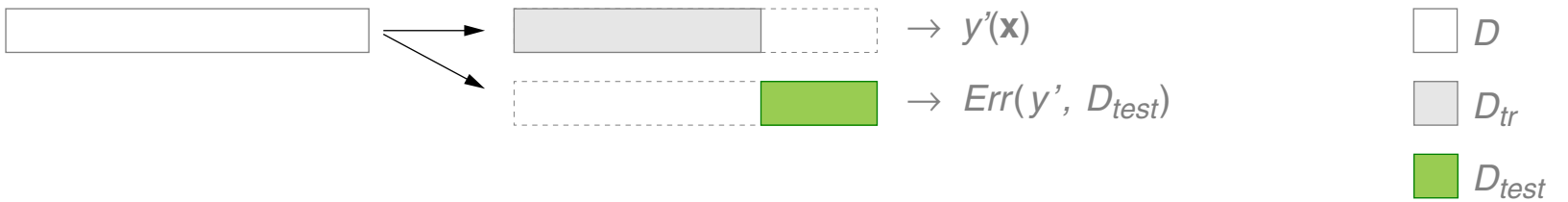
Evaluating Effectiveness

Holdout Error



Setting:

- $D_{test} \subset D$ is the test set.
- $y(\mathbf{x})$ is the classifier trained on D.
- $y'(\mathbf{x})$ is the classifier trained on $D_{tr} = D \setminus D_{test}$.

Holdout error of $y(\mathbf{x})$:

- $$\mathit{Err}(y, D_{test}) = \frac{|\{(\mathbf{x}, c(\mathbf{x})) \in D_{test} : y'(\mathbf{x}) \neq c(\mathbf{x})\}|}{|D_{test}|}$$

  $= \text{misclassification rate of } y'(\mathbf{x}) \text{ on the test set.}$

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Remarks:

- When splitting D into D_{tr} and D_{test} one has to ensure that the underlying distribution is maintained, i.e., the examples have to be drawn independently and according to P. If this condition is not fulfilled then $Err(y, D_{test})$ cannot be used as an estimation of $Err^*(y)$.
  Keywords: stratification, sample selection bias

- D_{tr} and D_{test} should have similar sizes. A typical value for splitting D into training set D_{tr} and test set D_{test} is 2:1.

- The fact that random variables are both independent of each other and identically distributed is often abbreviated to "i.i.d."

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Evaluating Effectiveness

k-Fold Cross-Validation

Setting:

- k test sets D_{test_i} by splitting D into k disjoint sets of similar size.

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Evaluating Effectiveness

k-Fold Cross-Validation

Setting:

- k test sets D_{test_i} by splitting D into k disjoint sets of similar size.
- $i = 1, \ldots, k$:

  $y'_i(\mathbf{x})$ is the classifier trained on $D_{tr} = D \setminus D_{test_i}$,

  $$Err(y'_i, D_{test_i}) = \frac{|\{(\mathbf{x}, c(\mathbf{x})) \in D_{test_i} : y'_i(\mathbf{x}) \neq c(\mathbf{x})\}|}{|D_{test_i}|}$$

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Evaluating Effectiveness

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Cross-validation error of $y(\mathbf{x})$:

- $$Err_{cv}(y, k) = \frac{1}{k} \sum_{i=1}^{k} Err(y'_i, D_{test_i})$$

  $=$ averaged misclassification rate on the k test sets.

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Remarks:

- For large k the set $D_{tr} = D \setminus D_{test_i}$ is of similar size as D. Hence $Err(y_i, D_{test_i})$—as well as $Err_{cv}(y, k)$—is close to $Err^*(y)$, since $Err^*(y)$ is the error of the classifier y trained on D.
- n-fold cross-validation (aka "leave one out") is the special case with $k = n$. Obviously singleton test sets ($|D_{test_i}| = 1$) are never stratified since they contain a single class only.
- n-fold cross-validation is a special case of exhaustive cross-validation methods, which learn and test on all possible ways to divide the original sample into a training and a validation set. [Wikipedia]

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Evaluating Effectiveness

Comparing Model Variants

Classifier training:

- $D = \{(\mathbf{x}_1, c(\mathbf{x}_1)), \ldots, (\mathbf{x}_n, c(\mathbf{x}_n))\} \subseteq X \times C$ is a set of examples.
- m hyperparameter values $\pi_1, \pi_2, \ldots, \pi_m$,

  m classifiers $y_{\pi_1}(\mathbf{x}), y_{\pi_2}(\mathbf{x}), \ldots, y_{\pi_m}(\mathbf{x})$.
- $y_{\pi_j}(\mathbf{x})$ is the classifier trained on D.
- Which is the most effective among the m classifiers $y_{\pi_j}(\mathbf{x})$?

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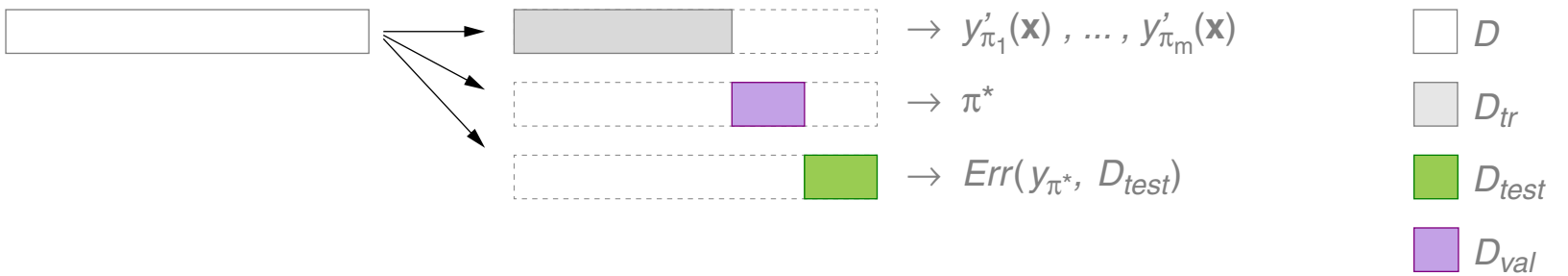
Evaluating Effectiveness

Model Selection (single validation set)



Setting:

- $D_{test} \subset D$ is the test set,

  $D_{val} \subset (D \setminus D_{test})$ is the validation set.

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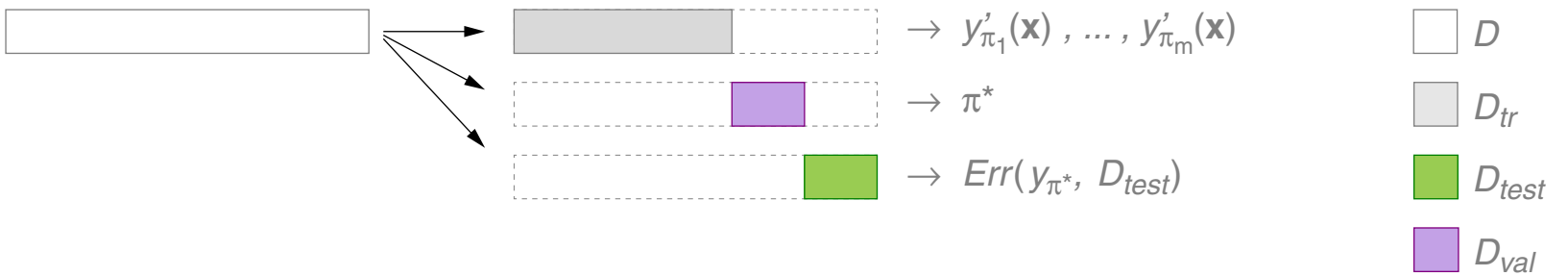
Evaluating Effectiveness

Model Selection (single validation set)



Setting:

- $D_{test} \subset D$ is the test set,

  $D_{val} \subset (D \setminus D_{test})$ is the validation set.

- $j = 1, \ldots, m$:

  $y'_{\pi_j}(\mathbf{x})$ is the classifier trained on $D_{tr} = D \setminus (D_{test} \cup D_{val})$,

$$\underline{Err(y'_{\pi_j}, D_{val})} = \frac{|\{(\mathbf{x}, c(\mathbf{x})) \in D_{val} : y'_{\pi_j}(\mathbf{x}) \neq c(\mathbf{x})\}|}{|D_{val}|}$$

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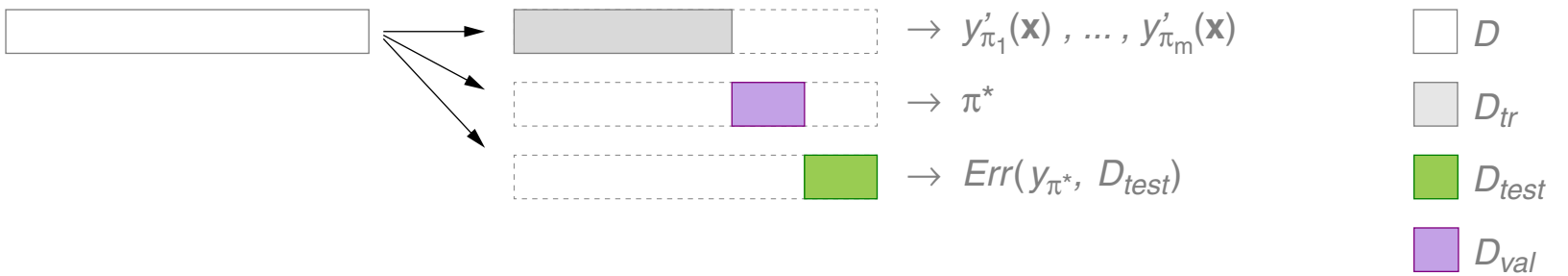
Evaluating Effectiveness

Model Selection (single validation set) [Holdout Error]



(continued)

- $\pi^* = \underset{\pi_j,\ j=1,\ldots,m}{\text{argmin}} \ \ Err(y'_{\pi_j}, D_{val})$
- $y_{\pi^*}(\mathbf{x})$ is the classifier trained on D,

  $y'_{\pi^*}(\mathbf{x})$ is the classifier trained on $D_{tr} = D \setminus D_{test}$.

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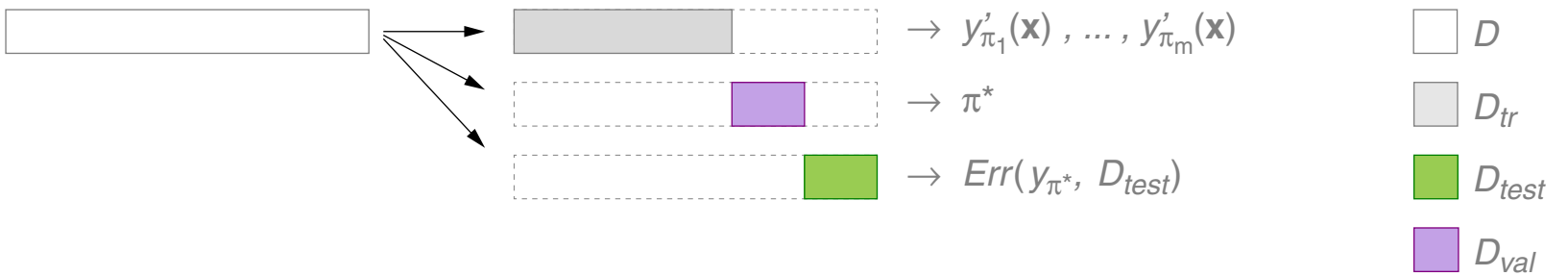
Evaluating Effectiveness

Model Selection (single validation set) [Holdout Error]



(continued)

- $\pi^* = \underset{\pi_j,\ j=1,\ldots,m}{\text{argmin}} \ \mathit{Err}(y'_{\pi_j}, D_{\mathit{val}})$
- $y_{\pi^*}(\mathbf{x})$ is the classifier trained on D,

  $y'_{\pi^*}(\mathbf{x})$ is the classifier trained on $D_{\mathit{tr}} = D \setminus D_{\mathit{test}}$.

Holdout error of $y_{\pi^*}(\mathbf{x})$:

- $\mathit{Err}(y_{\pi^*}, D_{\mathit{test}}) = \dfrac{|\{(\mathbf{x}, c(\mathbf{x})) \in D_{\mathit{test}} : y'_{\pi^*}(\mathbf{x}) \neq c(\mathbf{x})\}|}{|D_{\mathit{test}}|}$

  $=$ misclassification rate of $y'_{\pi^*}(\mathbf{x})$ on the test set.

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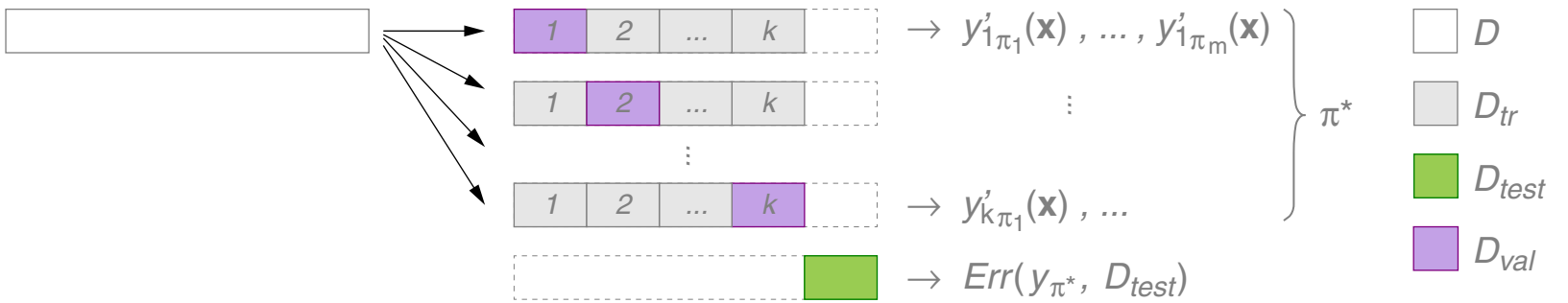
Evaluating Effectiveness

Model Selection (k validation sets)



Setting:

- $D_{test} \subset D$ is the test set,

  k validation sets D_{val_i} by splitting $D \setminus D_{test}$ into k disjoint sets of similar size.

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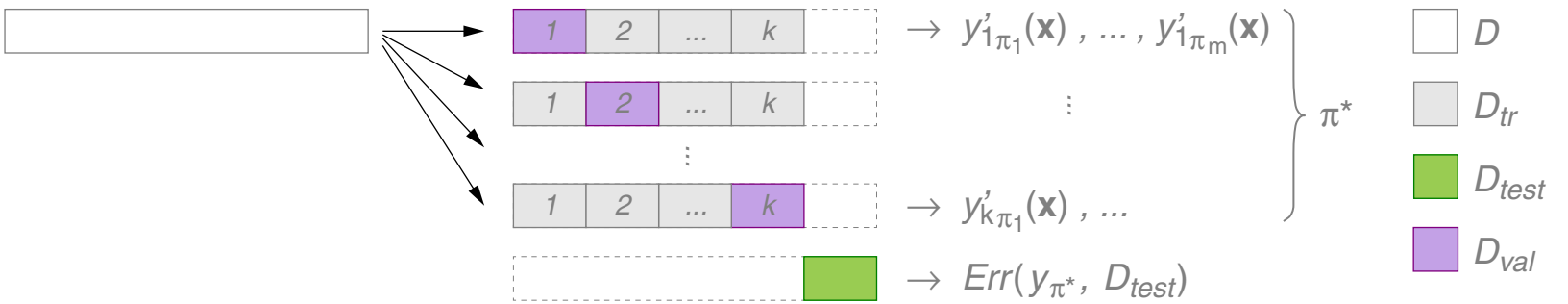
Evaluating Effectiveness

Model Selection (k validation sets)



Setting:

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- $i = 1, \ldots, k$:

  $j = 1, \ldots, m$:

  $y'_{i_{\pi_j}}(\mathbf{x})$ is the classifier trained on $D_{tr} = D \setminus (D_{test} \cup D_{val_i})$,

$$Err(y'_{i_{\pi_j}}, D_{val_i}) = \frac{|\{(\mathbf{x}, c(\mathbf{x})) \in D_{val_i} : y'_{i_{\pi_j}}(\mathbf{x}) \neq c(\mathbf{x})\}|}{|D_{val_i}|}$$

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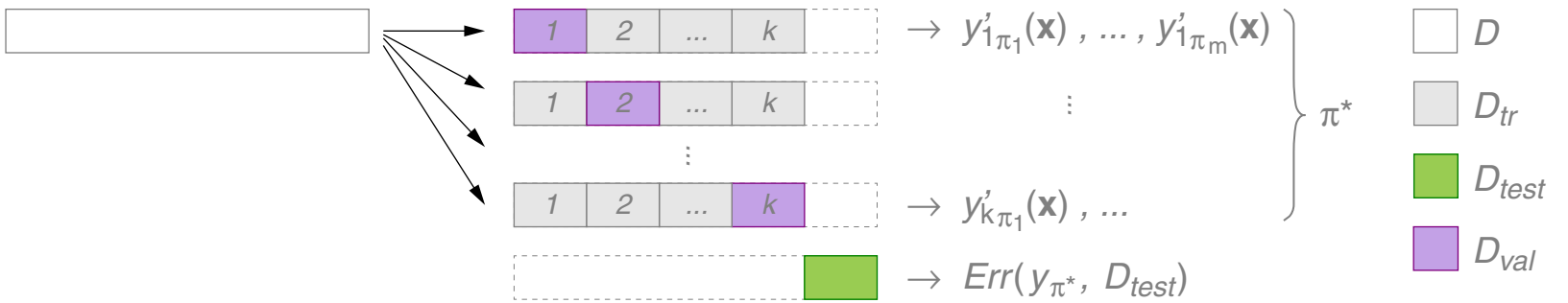
Evaluating Effectiveness

Model Selection (k validation sets)



(continued)

- $\pi^* = \underset{\pi_j,\ j=1,\ldots,m}{\text{argmin}} \sum_{i=1}^{k} Err(y'_{i_{\pi_j}}, D_{val_i})$

- $y_{\pi^*}(\mathbf{x})$ is the classifier trained on D,

  $y'_{\pi^*}(\mathbf{x})$ is the classifier trained on $D_{tr} = D \setminus D_{test}$.

Holdout error of $y_{\pi^*}(\mathbf{x})$: computation as before.

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Remarks:

- The validation set is also called “development set”.

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Evaluating Effectiveness

Misclassification Costs

Use of a *cost measure* for the misclassification of a feature vector $\mathbf{x} \in X$ in a wrong class c' instead of in the correct class c:

$$\textit{cost}(c' \mid c) \begin{cases} \geq 0 & \text{if } c' \neq c \\ = 0 & \text{otherwise} \end{cases}$$

Holdout error of y based on misclassification costs:

$$\textit{Err}_{\textit{cost}}(y, D_{\textit{test}}) = \frac{1}{|D_{\textit{test}}|} \cdot \sum_{(\mathbf{x}, c(\mathbf{x})) \in D_{\textit{test}}} \textit{cost}(y(\mathbf{x}) \mid c(\mathbf{x}))$$

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Remarks:

- The true error, $\mathit{Err}^*(y)$, is a special case of $\mathit{Err}_{\mathit{cost}}(y)$ with $\mathit{cost}(c' \mid c) = 1$ for $c' \neq c$. Consider in this regard the notation of $\mathit{Err}^*(y)$ in terms of the function $I(y(\mathbf{x}), c)$:

$$\mathit{Err}^*(y) \;=\; \frac{|\{\mathbf{x} \in X : y(\mathbf{x}) \neq c(\mathbf{x})\}|}{|X|} \;=\; \sum_{\mathbf{x} \in X} I(y(\mathbf{x}), c)$$

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